Technical University of Crete
School of Electrical and Computer Engineering
Course: Reinforcement Learning and Dynamic Optimization
Programming Assignment 1 : Stochastic Bandits
Report Submission Date: March 8, 2026
Instructor: Thrasyvoulos Spyropoulos

Areti Xynompiti

Overview

We consider a stochastic Multi-Armed Bandit problem arising in a wireless communication scenario. Our goal is for our algorithm to learn the best channel allocation for two users over time while maximizing the total transmission rate over 5 different channels.

UCB Implementation for Case A

In this case at each round t, the learner selects one arm and the only available info is the reward $X_{i_t,t}$ corresponding to the total transmission rate of both users. The goal of the algorithm is therefore to estimate the expected reward for every arm i and identify the optimal arm.

1 Theoretical Analysis

1.1 UCB index

To construct the UCB of each arm, we apply Hoeffding's inequality:

$$\mathbb{P}\Big(|\hat{\mu} - \mu| > \epsilon\Big) \leq \exp\left(\frac{-2N\epsilon^2}{(b-a)^2}\right)$$

When arm i is played, the reward is $X_{i,t} = \log(1+10\, s_{1,j_1,t}) + \log(1+10\, s_{2,j_2,t})$, where $s_{u,j,t} \sim \text{Unif}[s_{u,j} - \epsilon,\ s_{u,j} + \epsilon]$ with $s_{u,j} \in [0.8, 2.0]$ and $\epsilon = 0.6$, hence $s_{u,j,t} \in [0.2, 2.6]$. For a single user the instantaneous rate is $R = \log(1+10s)$, which leads to the range $R \in [\log 3, \log 27]$. Since the total reward is the sum of the two user rates, we obtain $X_{i,t} \in [a, b]$ with $b - a = 2\log 9$. Substituting this into Hoeffding's inequality gives

$$\mathbb{P}\Big(Bad_i\Big) = \mathbb{P}\Big(|\hat{\mu}_i - \mu_i| > \epsilon\Big) \leq \exp\left(-\frac{2N_i\epsilon^2}{(2\log 9)^2}\right)$$

To ensure the bad event "almost never happens" , which allows us to proceed to regret proof assuming the Good Event holds, we choose $\exp\Big(-\frac{2N_i\epsilon^2}{(2\log 9)^2}\Big) = \frac{1}{T^4} \ \Rightarrow\ \epsilon = (2\log 9)\sqrt{\frac{2\log T}{N_i(t)}}$.

Thus the UCB index used by the algorithm is $\text{UCB}_i(t) = \hat{\mu}_i(t) + (2\log 9)\sqrt{\frac{2\log T}{N_i(t)}}$

1.2 Instance-dependent regret bound

We start by assuming that the Good Event holds, $|\hat{\mu}_i(t) - \mu_i| \leq (2\log 9)\sqrt{\frac{2\log T}{N_i(t)}}$, and since we are trying to upper bound we will use the right inequality : $\hat{\mu}_i(t) \leq \mu_i + (2\log 9)\sqrt{\frac{2\log T}{N_i(t)}}$ (1)
Lets say arm i ,which is suboptimal, gets selected ,then: $\hat{\mu}_i(t) + (2\log 9)\sqrt{\frac{2logT}{N_i(t)}} \geq \hat{\mu}_*(t) + (2\log 9)\sqrt{\frac{2\log T}{N_*(t)}}$ (2)
We return to relation (1) and add the term ϵ in both sides in order to form the $\text{UCB}_i(t)$ term :

$$\mu_i + (4\log 9)\sqrt{\frac{2\log T}{N_i(t)}} \geq \hat{\mu}_i(t) + (2\log 9)\sqrt{\frac{2\log T}{N_i(t)}} \geq \text{UCB} * (t), \text{due to equation (2)}$$

$$\Rightarrow \mu_i + (4\log 9)\sqrt{\frac{2\log T}{N_i(t)}} \geq \mu* \Rightarrow \mu * -\mu_i \leq (4\log 9)\sqrt{\frac{2\log T}{N_i(t)}} \Rightarrow \Delta_i \leq (4\log 9)\sqrt{\frac{2\log T}{N_i(t)}}$$

$$\Rightarrow \boxed{N_i(t) \leq \frac{8(2\log 9)^2 \log(T)}{\Delta_i^2}}$$

This shows that the number of pulls of any suboptimal arm grows logarithmically with the time horizon T, which leads to the instance-dependent regret bound of the UCB algorithm.
Since the bound on $N_i(T)$ holds under the Good Event, we write

$$\mathbb{E}[R(T) \mid Good] \leq \sum_{i:\Delta_i>0} \Delta_i \cdot \frac{8(2\log 9)^2 \log(T)}{\Delta_i^2} = \sum_{i:\Delta_i>0} \frac{8(2\log 9)^2 \log(T)}{\Delta_i},$$

We know that the mean regret is : $\mathbb{E}[R(T)] = \mathbb{E}[R(T) \mid Good]\mathbb{P}(Good) + \mathbb{E}[R(T) \mid Bad]\mathbb{P}(Bad)$, and since previously we claimed the bad events "almost never" happen then we can say that the dominant term is the "Good" one ,leading to : $\boxed{\mathbb{E}[R(T)] \leq \sum_{i:\Delta_i>0} \frac{8(2\log 9)^2 \log(T)}{\Delta_i}}$

2 Simulation Setup

2.1 Specific Environment

The baseline channel quality matrix $S = [s_{u,j}]$ describes the quality for each user–channel pair and remains fixed throughout the implementation, representing the "average" user–channel quality, while the values $s_{u,j,t}$ used in the implementation represent the instantaneous SNR obtained from fluctuations around this mean according to $s_{u,j,t} \sim \text{Unif}[s_{u,j} - \epsilon,\ s_{u,j} + \epsilon]$.

$$S = \begin{bmatrix} 1.8867 & 0.8927 & 1.1271 & 1.5462 & 1.9208 \\ 0.9019 & 1.5551 & 1.7307 & 1.4137 & 1.6982 \end{bmatrix}$$

2.2 Arm Definition and Mean Rewards

The pair of channel combinations (j_1, j_2) for user 1 and user 2 $(K * (K-1))$ constitute the arms of our problem, and the reward we gain each time t an arm is "played" is given by $X_{i,t}$.
Regarding the calculation of the mean reward of each arm we know that we must have:
$\mu_i = \mathbb{E}[X_{i,n}] = \mathbb{E}[\log(1 + 10\, s_{1,j_1,t}) + \log(1 + 10\, s_{2,j_2,t})])$, due to linearity of expectation $\Rightarrow$
$\mu_i = \mathbb{E}[\log 1 + 10\, s_{1,j_1,t}] + \mathbb{E}[\log 1 + 10\, s_{2,j_1,t}]$

Let $g(x) = \log(1+10\, x)$, where x follows uniform distibution . Then the pdf is: $f_{s_{u,j}}(x) = \begin{cases} \dfrac{1}{2\epsilon}, & x \in [s_{u,j} - \epsilon,\ s_{u,j} + \epsilon] \\ 0, & \text{otherwise} \end{cases}$

Therefore, $\mathbb{E}[g(s_{u,j,t})] = \int_{s_{u,j}-\epsilon}^{s_{u,j}+\epsilon} g(x)\, f_{s_{u,j}}(x)\, dx \Rightarrow \mathbb{E}[\log(1 + 10\, s_{u,j,t})] = \frac{1}{2\epsilon} \int_{s_{u,j}-\epsilon}^{s_{u,j}+\epsilon} \log(1 + 10x)\, dx$
To compute $\int \log(1+10x)\, dx$ we apply integration by parts and obtain $\int \log(1+10x)\, dx = \frac{(1+10x)\log(1+10x)-(1+10x)}{10}$
So the anti-derivative is $F(x) = \frac{(1+10x)\log(1+10x)-(1+10x)}{10}$
Therefore by applying the limits f integration we obtain: $\int_a^b \log(1 + 10x)\, dx = F(b) - F(a)$
Substituting this into the expression for μ_i, we get $\mu_i = \frac{1}{2\epsilon}\left[F(s_{1,j_1} + \epsilon) - F(s_{1,j_1} - \epsilon)\right] + \frac{1}{2\epsilon}\left[F(s_{2,j_2} + \epsilon) - F(s_{2,j_2} - \epsilon)\right]$

UCB Implementation for Case B

In this case the learner selects a pair of channels (j_1, j_2) and observes the two individual rewards $X_{1,j_1,t}$ and $X_{2,j_2,t}$,so the learner has access to the reward of each user separately. Therefore, the expected reward of an arm decomposes into two components: $\mu_{j_1,j_2} = \mu_{1,j_1} + \mu_{2,j_2}$. This allows us to split the problem into two independent bandit problems, one for each user. Each user now must learn the best channel among the K available ones, so instead of treating all $K * (K-1)$ channel pairs as independent arms, we maintain two separate UCB learners, each operating over K arms. At every round the algorithm selects the channel with the highest UCB index for each user independently, and the selected pair (j_1, j_2) is what we "play".

Theoretical Analysis

1.1 UCB index for Case B

In this case, the reward is treated separately for each user. Hence, for user 1 and channel j_1 we define $X_{1,j_1,t} = \log(1 + 10\, s_{1,j_1,t})$, similarly for user 2 and channel j_2.
Thus, for a single user, $X_{u,j,t} = \log(1 + 10\, s_{u,j,t}) \in [\log 3, \log 27]$, so for each user separately, $b - a = \log 9$.

1.2 Instance-dependent regret bound

The expected reward of arm (j_1, j_2) is: $\mu_{j_1,j_2} = \mu_{1,j_1} + \mu_{2,j_2}$. Therefore the UCB index can be written as:

$$\mathrm{UCB}_{j_1,j_2}(t) = \hat{\mu}_{1,j_1}(t) + (\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}} + \hat{\mu}_{2,j_2}(t) + (\log 9)\sqrt{\frac{2\log T}{N_{2,j_2}(t)}}.$$

The maximization over all arms decomposes into two independent maximizations, one for each user:

$$j_{1,t} = \arg\max_{j_1}\left(\hat{\mu}_{1,j_1}(t) + (\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}}\right), j_{2,t} = \arg\max_{j_2}\left(\hat{\mu}_{2,j_2}(t) + (\log 9)\sqrt{\frac{2\log T}{N_{2,j_2}(t)}}\right).$$

Therefore each user can be analyzed as an independent UCB problem:
We start by assuming that the Good Event holds for User 1: $|\hat{\mu}_{1,j_1}(t) - \mu_{1,j_1}| \leq (\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}}$, and since we are trying to upper bound we use the right inequality: $\hat{\mu}_{1,j_1}(t) \leq \mu_{1,j_1} + (\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}}$ (1)
Lets say channel j_1, which is suboptimal for User 1, gets selected ,then:

$$\hat{\mu}_{1,j_1}(t) + (\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}} \geq \hat{\mu}_{1,j_1^*}(t) + (\log 9)\sqrt{\frac{2\log T}{N_{1,j_1^*}(t)}} \quad (2)$$

We return to relation (1) and add the ϵ term in both sides in order to form the $\mathrm{UCB}_{1,j_1}(t)$ term:

$$\mu_{1,j_1} + (2\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}} \geq \hat{\mu}_{1,j_1}(t) + (\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}} \geq \mathrm{UCB}_{1,j_1^*}(t), \text{ due to equation (2)}$$

Since the optimal index is always at least its mean, $\mathrm{UCB}_{1,j_1^*}(t) \geq \mu_{1,j_1^*}$, we obtain

$$\mu_{1,j_1} + (2\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}} \geq \mu_{1,j_1^*} \Rightarrow \mu_{1,j_1^*} - \mu_{1,j_1} \leq (2\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}} \Rightarrow \Delta_{1,j_1} \leq (2\log 9)\sqrt{\frac{2\log T}{N_{1,j_1}(t)}}$$

$$\Rightarrow \boxed{N_{1,j_1}(t) \leq \frac{8(\log 9)^2 \log(T)}{\Delta_{1,j_1}^2}}$$

Similarly for User 2, for any suboptimal channel j_2 we obtain: $\boxed{N_{2,j_2}(t) \leq \frac{8(\log 9)^2 \log(T)}{\Delta_{2,j_2}^2}}$
The instantaneous regret is: $\mu^* - \mu_{j_{1,t},j_{2,t}} = (\mu_1^* - \mu_{1,j_{1,t}}) + (\mu_2^* - \mu_{2,j_{2,t}}) = \Delta_{1,j_{1,t}} + \Delta_{2,j_{2,t}}$
Therefore the cumulative regret becomes:

$$R(T) = \sum_{t=1}^{T} \left(\Delta_{1,j_{1,t}} + \Delta_{2,j_{2,t}}\right) \Rightarrow R(T) = \sum_{j_1 \neq j_1^*} \Delta_{1,j_1} N_{1,j_1}(T) + \sum_{j_2 \neq j_2^*} \Delta_{2,j_2} N_{2,j_2}(T).$$

So given the Good event the we write:

$$\mathbb{E}[R(T) \mid Good] \leq \sum_{j_1 \neq j_1^*} \frac{8(\log 9)^2 \log(T)}{\Delta_{1,j_1}} + \sum_{j_2 \neq j_2^*} \frac{8(\log 9)^2 \log(T)}{\Delta_{2,j_2}}$$

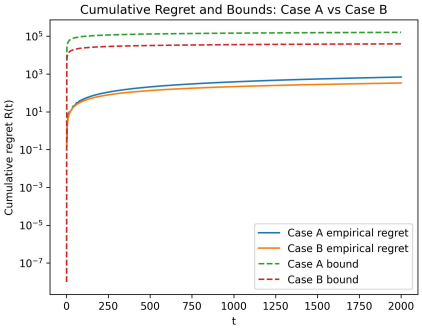
As in Case A, the bad events happen with very small probability, the dominant term is again the Good Event term, leading to $\boxed{\mathbb{E}[R(T)] \leq \sum_{j_1 \neq j_1^*} \frac{8(2\log 9)^2 \log(T)}{\Delta_{1,j_1}} + \sum_{j_2 \neq j_2^*} \frac{8(2\log 9)^2 \log(T)}{\Delta_{2,j_2}}}$

Thus the regret sum runs over the suboptimal channels of each user, which is $2(K-1)$ terms, instead of the $K(K-1)$ suboptimal channel pairs appearing in Case A, which leads to a smaller bound.

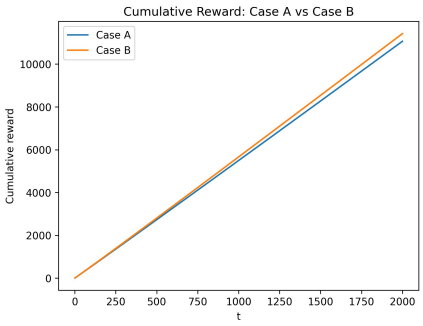
Results



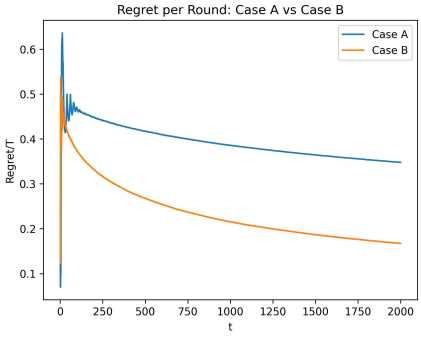
(a) Cumulative regret



(b) Regret with bounds



(c) Cumulative reward



(d) Regret per round

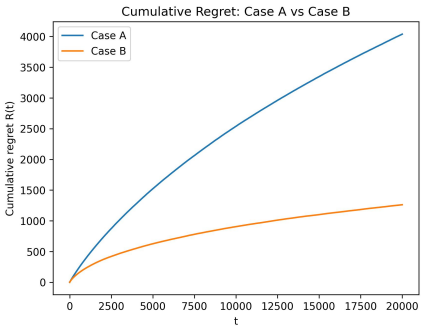
From the cumulative regret plots we can see that the regret follows a logarithmic trend, which means that the algorithm indeed learns over time.

Comparing the two cases, regret-per-round plot shows the same behavior as the regret of Case B decreases faster and stays lower than the regret of Case A ,which was expected given the analysis above.

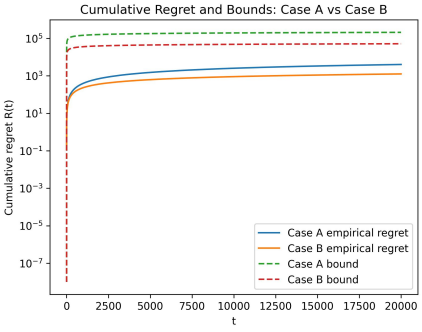
Looking now at the cumulative reward, we see that Case B collects slightly more total reward than Case A, which is expected because lower regret means that the algorithm selects better arms more often.

Finally, the plots with the theoretical bounds show that the empirical regret stays well below the bound during the whole horizon, which agrees with the theory and confirms the logarithmic regret behavior predicted by the analysis of the UCB algorithm we did above.

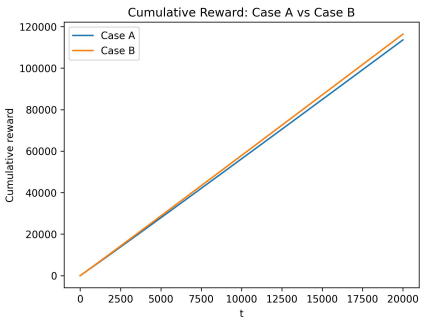
Figure 1: Comparison of Case A and Case B for $T = 2000$.



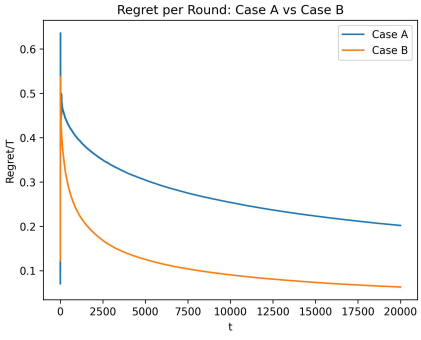
(a) Cumulative regret



(b) Regret with bounds



(c) Cumulative reward

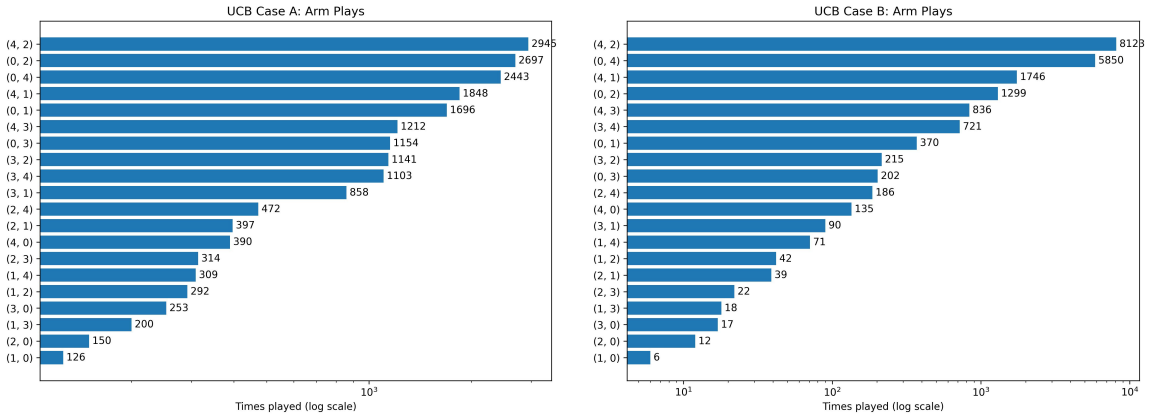


(d) Regret per round

For the larger horizon $T = 20000$, the regret per round becomes much smaller and even drops below 0.1. This shows that the algorithm has already identified the optimal arm in most rounds and only occasionally selects suboptimal arms.

Finally, although it might be difficult to clearly observe from the plot, we see that the cumulative regret of both cases gets closer to the theoretical bound. This suggests that if we run the algorithm for an even larger horizon, the empirical regret will approach the bound asymptotically, as predicted by the theoretical analysis.

Figure 2: Comparison of Case A and Case B for $T = 20000$.



(a) Arm plays for Case A

(b) Arm plays for Case B

Figure 3: Number of times each arm was selected for $T = 20000$.

From the plots we observe that some arms are played many times, such as $(0, 2)$ and $(0, 4)$, which means their expected rewards are close to the optimal. In Case A these arms are played almost as many times as the optimal arm. In contrast, in Case B they are played much fewer times while the optimal arm is selected significantly more often.

Notation

CASE A

K	number of available channels
T	time horizon (total number of rounds)
n	time index, $n = 1, \ldots, T$
$i = (j_1, j_2)$	arm corresponding to the pair of channels assigned to the two users
$X_{i,n}$	reward obtained when arm i is selected at time n
$\mu_i = \mathbb{E}[X_{i,n}]$	expected reward (true mean) of arm i
μ^*	expected reward of the optimal arm
$\Delta_i = \mu^* - \mu_i$	suboptimality gap of arm i
$N_i(T)$	number of times arm i is selected up to time T
$\mathbb{E}[R(T)] = \sum_{i=1}^{K^2} \Delta_i \, \mathbb{E}[N_i(T)]$	expected regret after T rounds
$s_{u,j}$	baseline channel quality for user u on channel j
$s_{u,j,t} \sim \text{Unif}[s_{u,j} - \epsilon,\ s_{u,j} + \epsilon]$	instantaneous channel quality at time t
$X_{i,t} = \log(1 + 10\, s_{1,j_1,t}) + \log(1 + 10\, s_{2,j_2,t})$	reward obtained when arm $i = (j_1, j_2)$ is selected at time t

CASE B

$X_{1,j_1,t} = \log(1 + 10\, s_{1,j_1,t})$	instantaneous reward of user 1 on channel j_1
$X_{2,j_2,t} = \log(1 + 10\, s_{2,j_2,t})$	instantaneous reward of user 2 on channel j_2
$\mu_{1,j_1} = \mathbb{E}[X_{1,j_1,t}]$	expected reward of user 1 on channel j_1
$\mu_{2,j_2} = \mathbb{E}[X_{2,j_2,t}]$	expected reward of user 2 on channel j_2
$\mu_{j_1,j_2} = \mu_{1,j_1} + \mu_{2,j_2}$	expected reward of arm (j_1, j_2)
$\Delta_{1,j_1} = \mu_1^* - \mu_{1,j_1}$	suboptimality gap for user 1 on channel j_1
$\Delta_{2,j_2} = \mu_2^* - \mu_{2,j_2}$	suboptimality gap for user 2 on channel j_2
$\Delta_{j_1,j_2} = \Delta_{1,j_1} + \Delta_{2,j_2}$	suboptimality gap of arm (j_1, j_2)